

Guana River Marsh Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

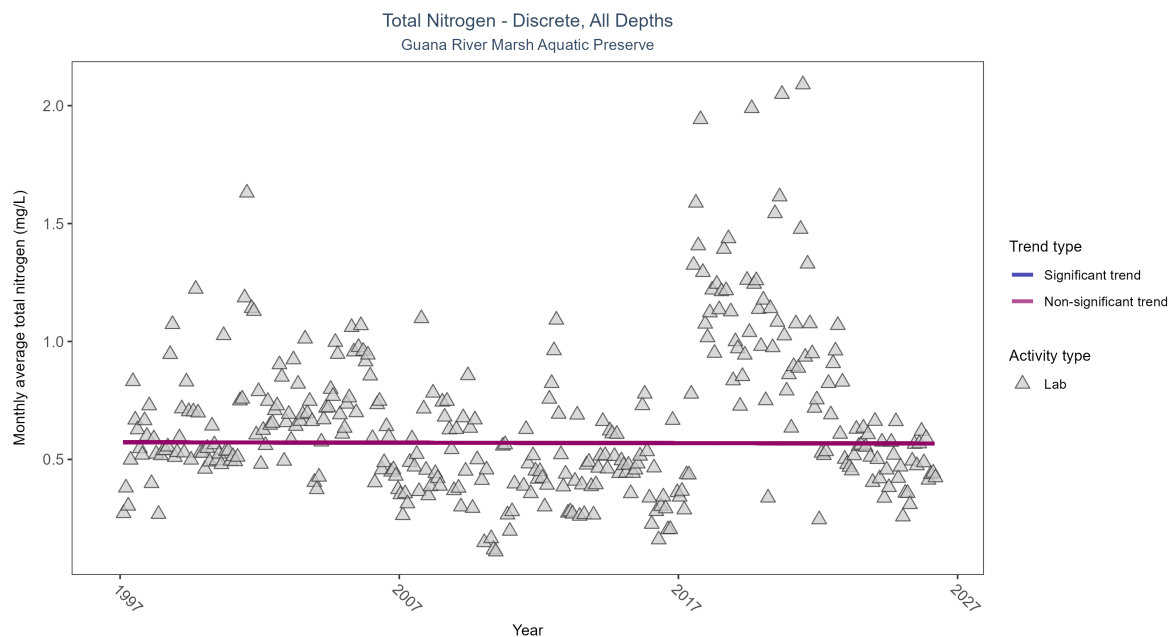


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Total nitrogen showed no detectable trend between 1997 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	2066	30	1997 - 2026	0.607	-0.00394	0.57257	-0.00018	0.9248



Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

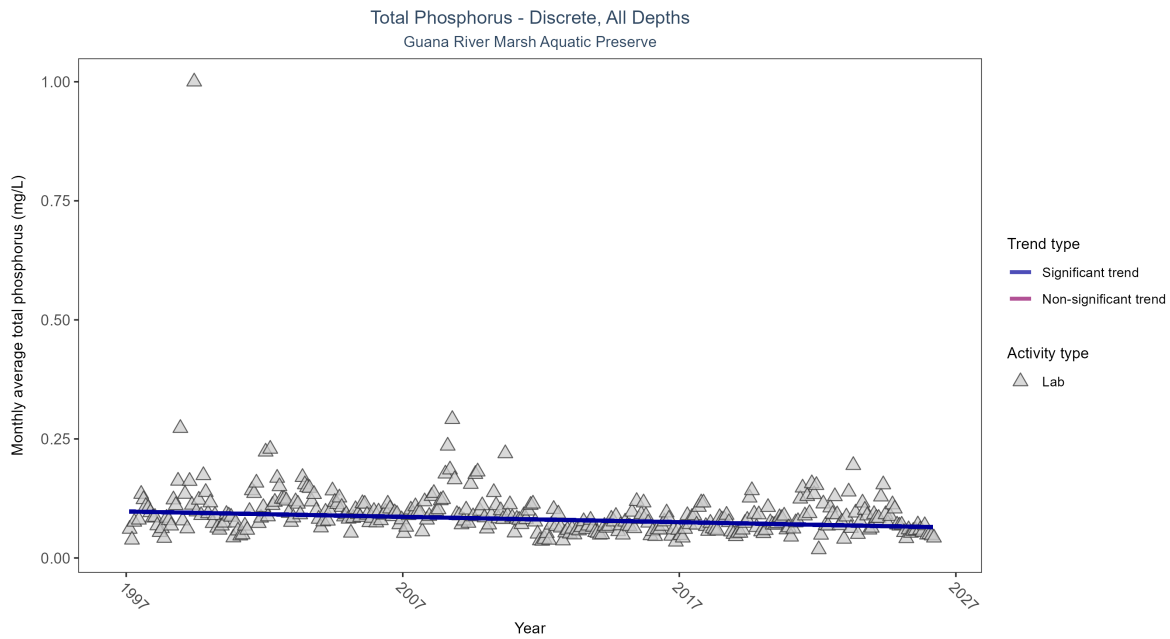


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	2996	30	1997 - 2026	0.075	-0.23835	0.09759	-0.00112	0



Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

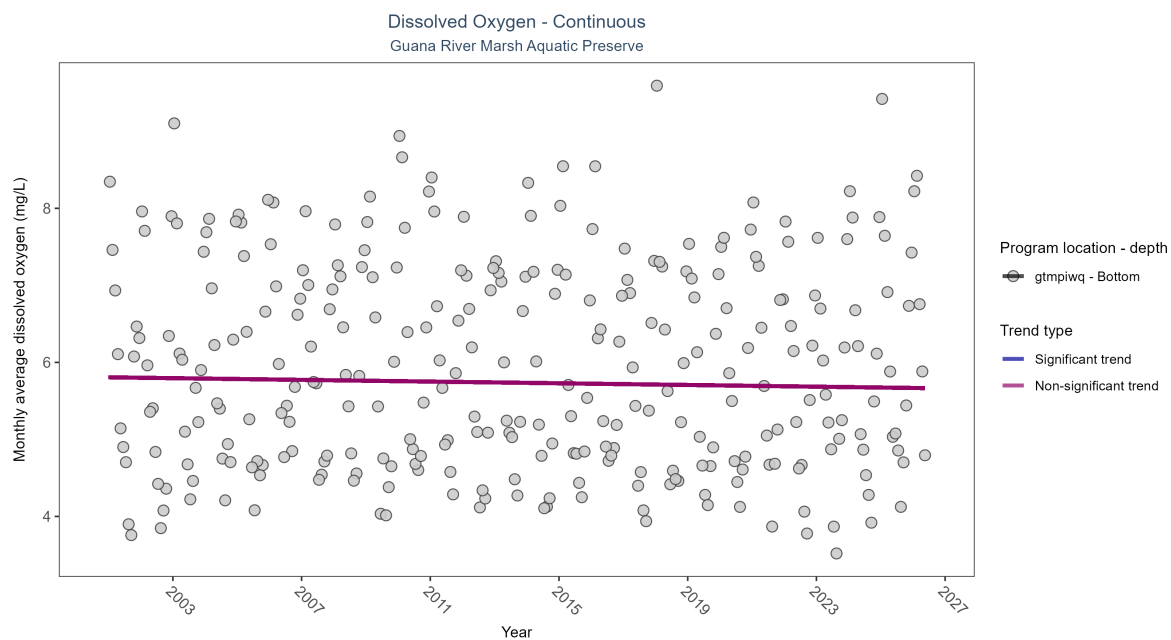


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: No detectable change in monthly average dissolved oxygen was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	No detectable trend	708889	26	2001 - 2026	5.9	-0.08	5.81	-0.01	0.07

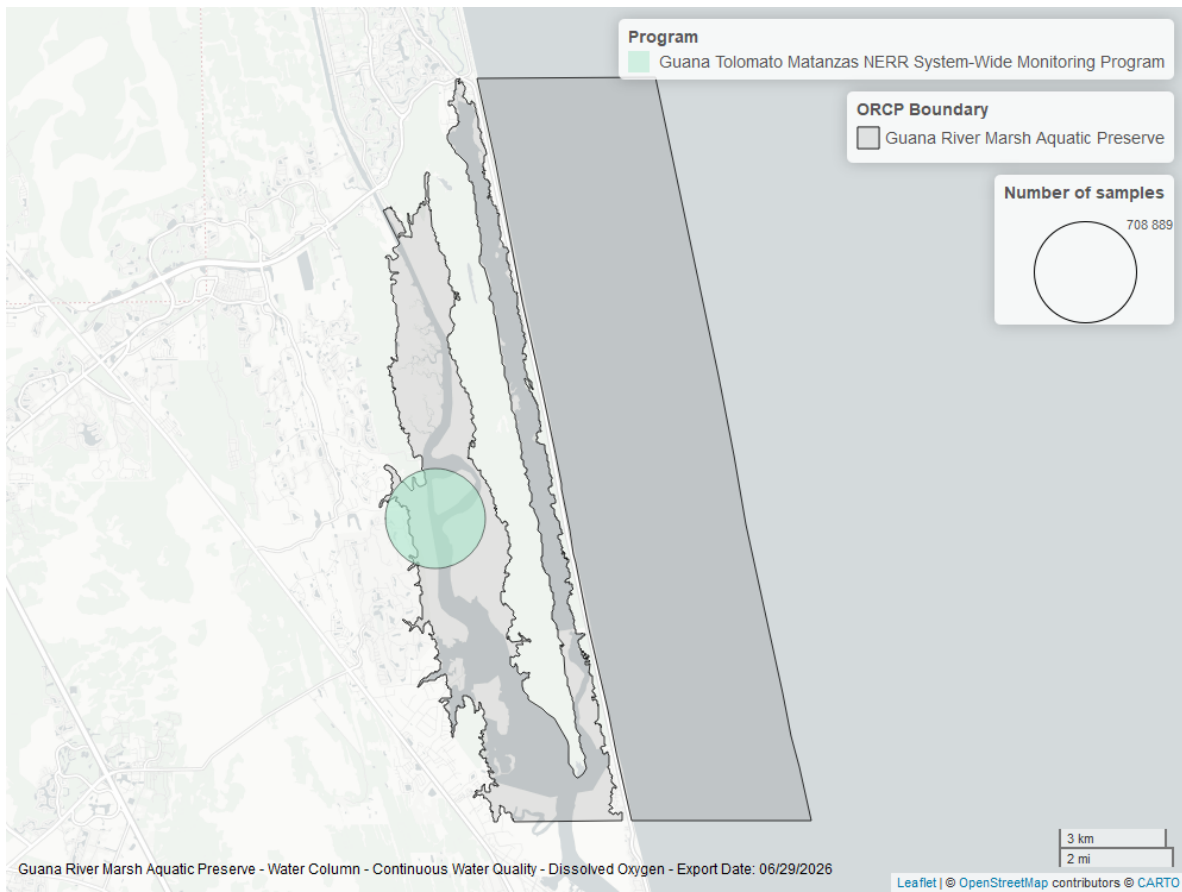


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

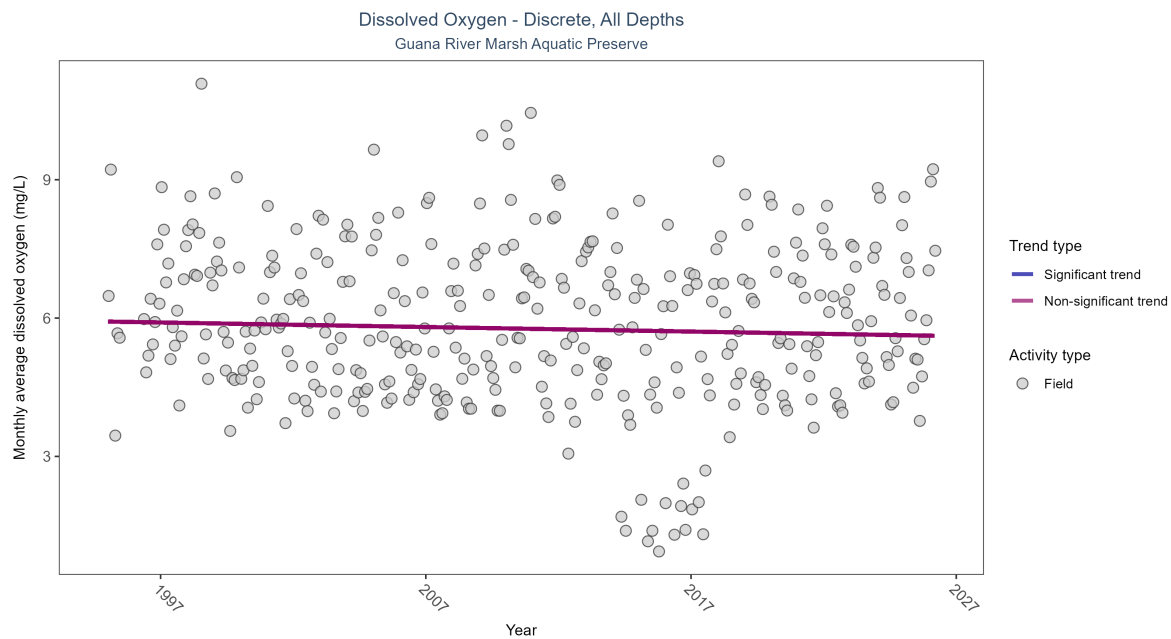


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Dissolved oxygen showed no detectable trend between 1995 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	7710	32	1995 - 2026	5.7	-0.06498	5.92403	-0.00985	0.082

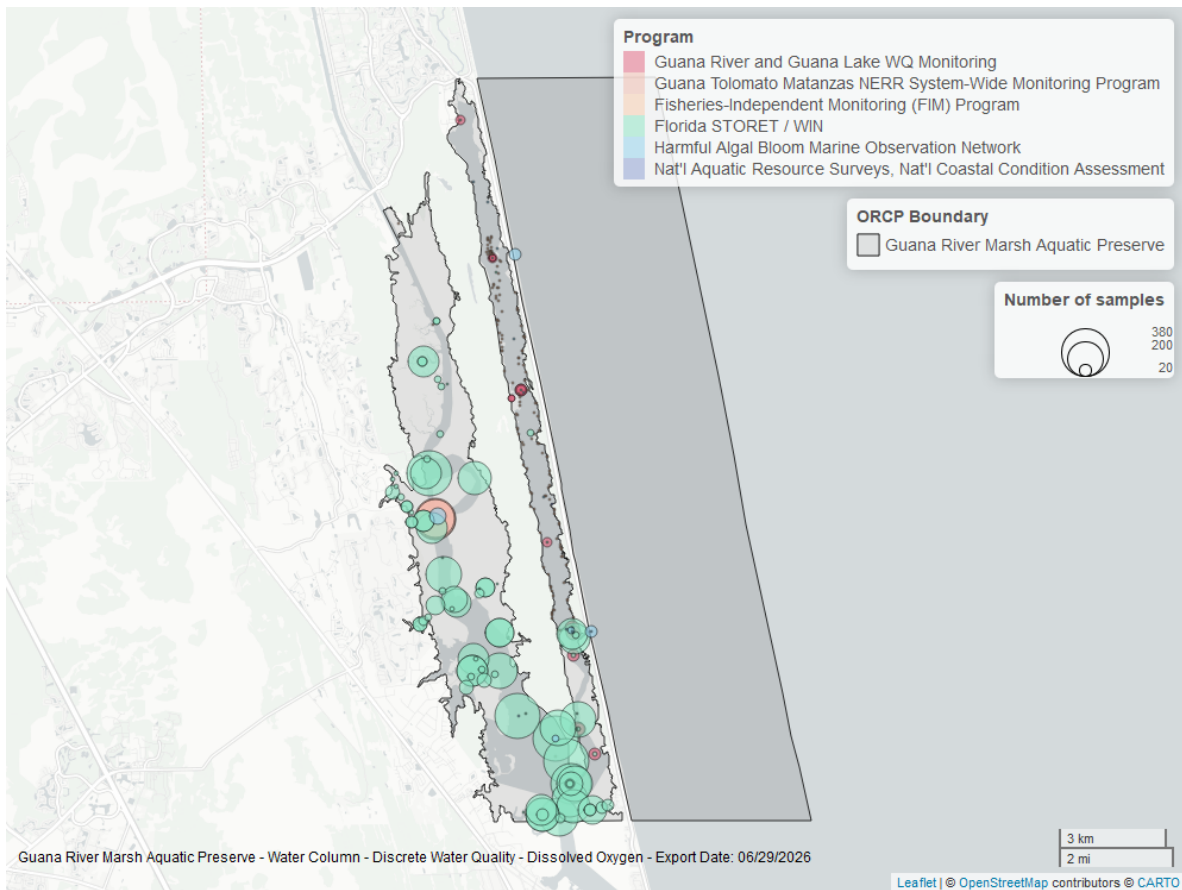


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

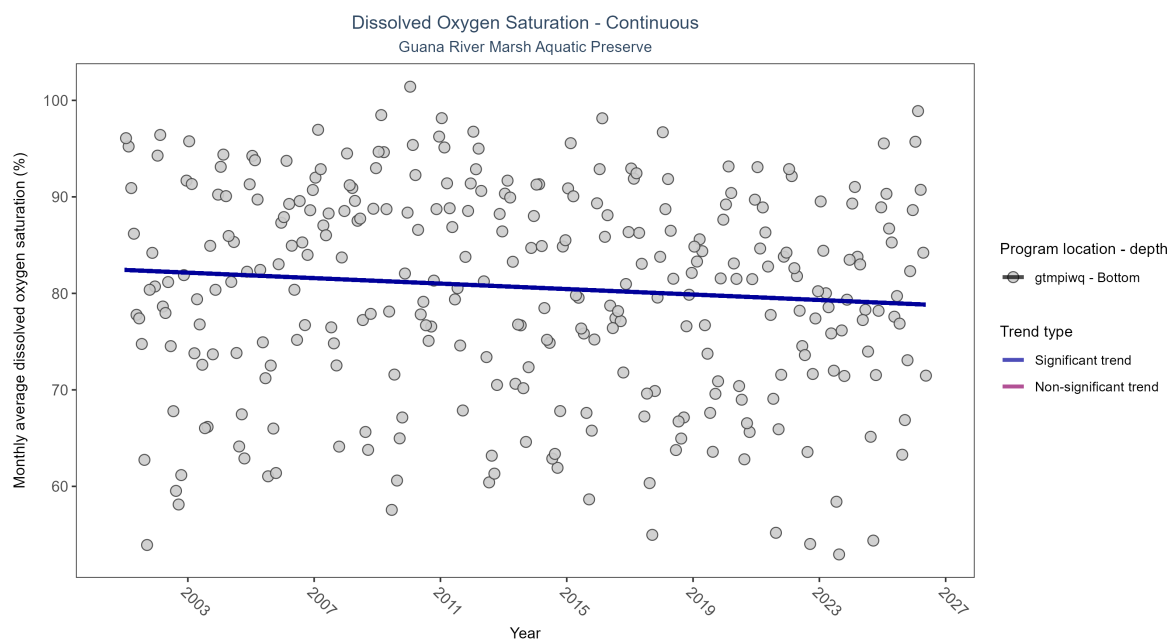


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

At one program location, monthly average dissolved oxygen saturation decreased by 0.14% per year.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Decreasing trend	714707	26	2001 - 2026	82.3	-0.15	82.43	-0.14	0

\end{table}

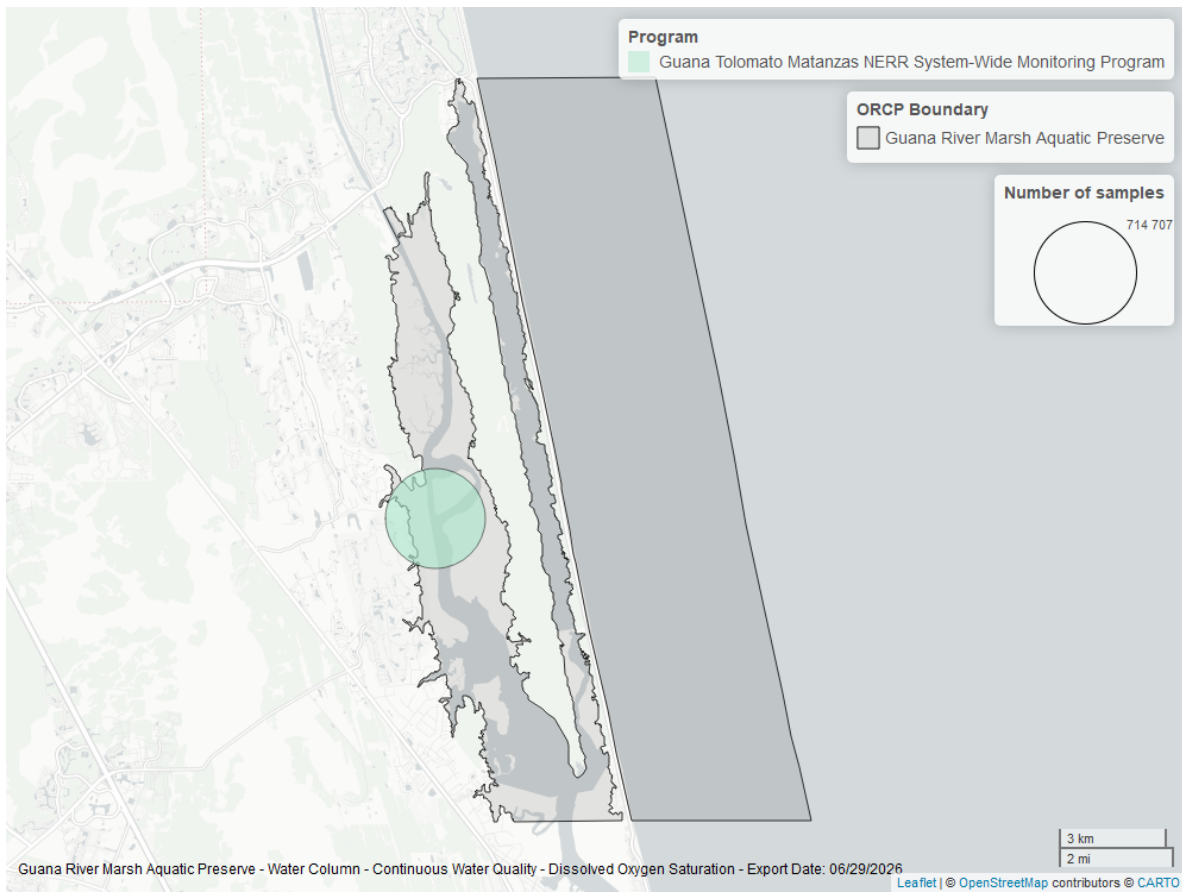


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

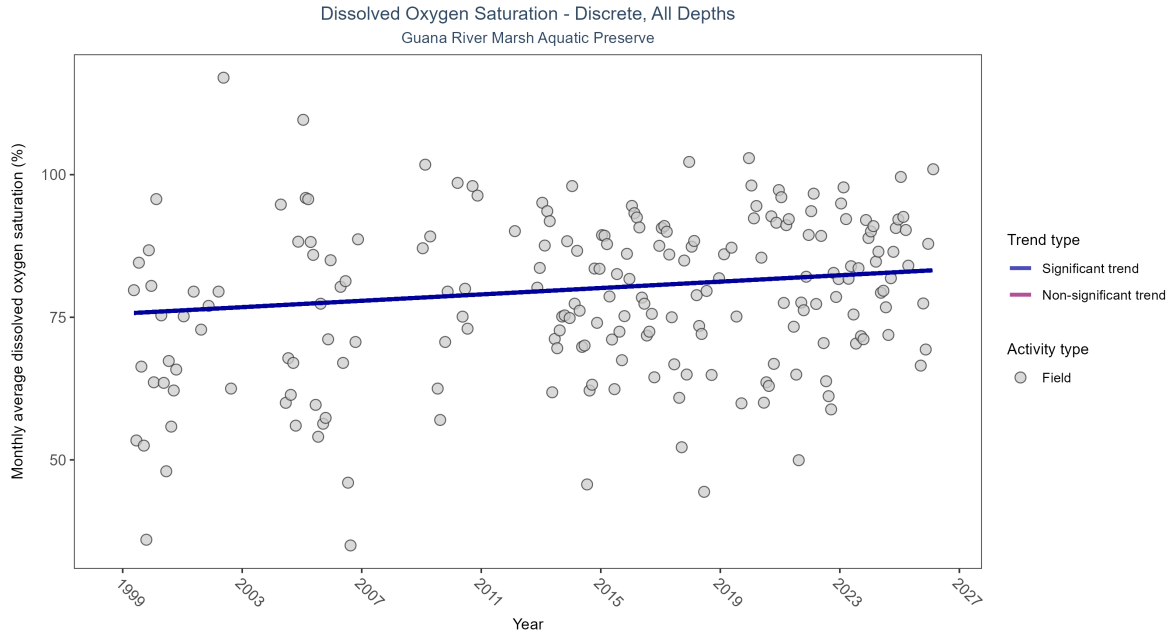


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

\begin{table}[H] \caption{Monthly average dissolved oxygen saturation increased by 0.28% per year.}

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	1231	24	1999 - 2026	79	0.14813	75.65186	0.27976	0.0038

\end{table}



Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

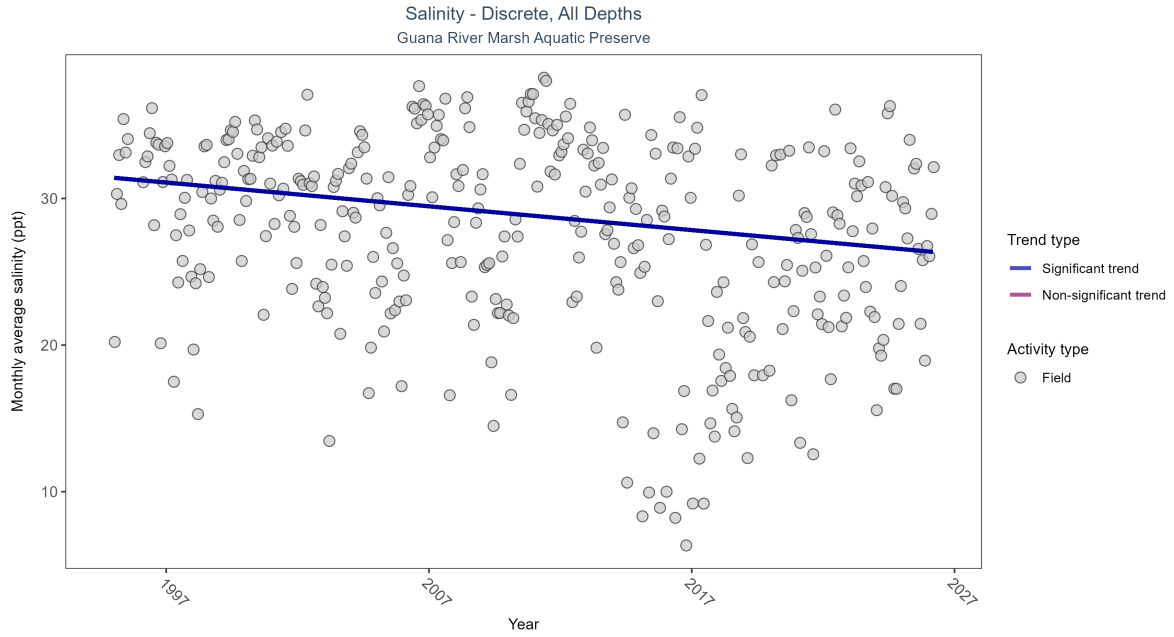


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 5: Monthly average salinity decreased by 0.16 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	8517	32	1995 - 2026	30.9	-0.18029	31.41319	-0.16222	0

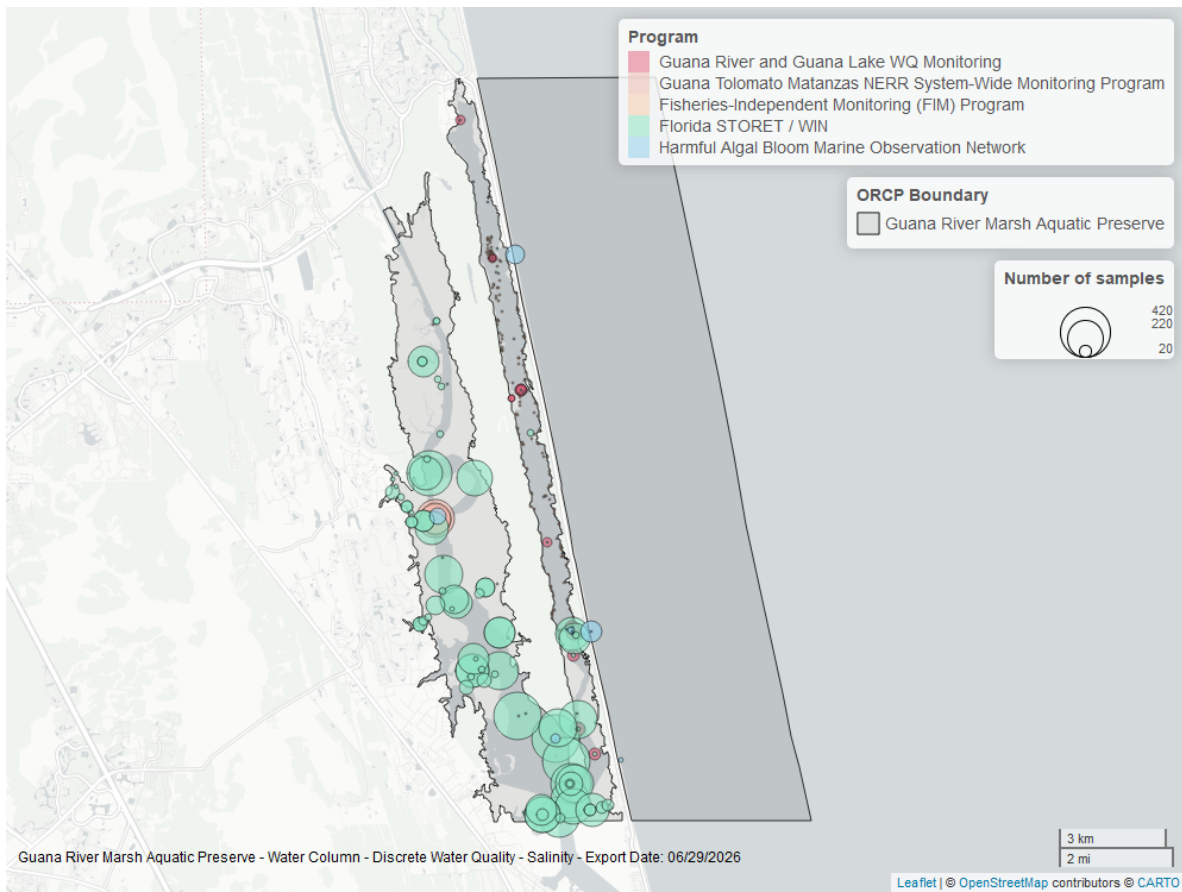


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

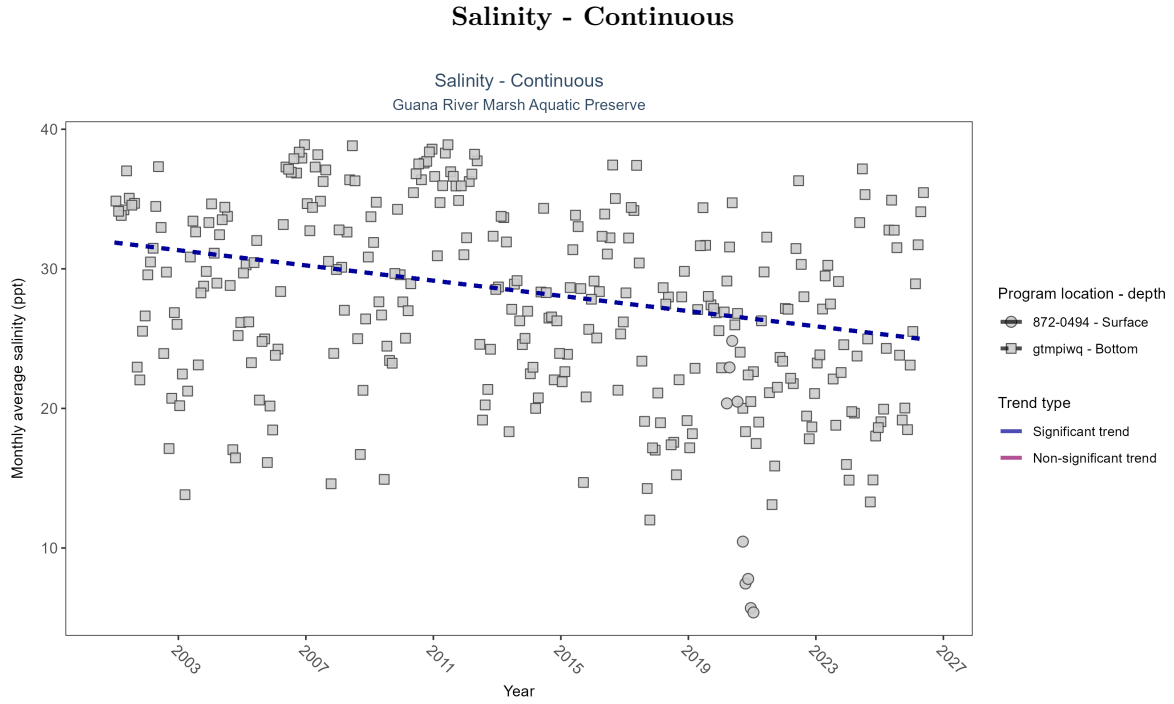


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 6: At one program location, monthly average salinity decreased by 0.27 ppt per year. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmptiwq	Decreasing trend	713006	26	2001 - 2026	27.80	-0.25	31.88	-0.27	0
872-0494	Insufficient data	34918	2	2020 - 2021	8.99	-	-	-	-

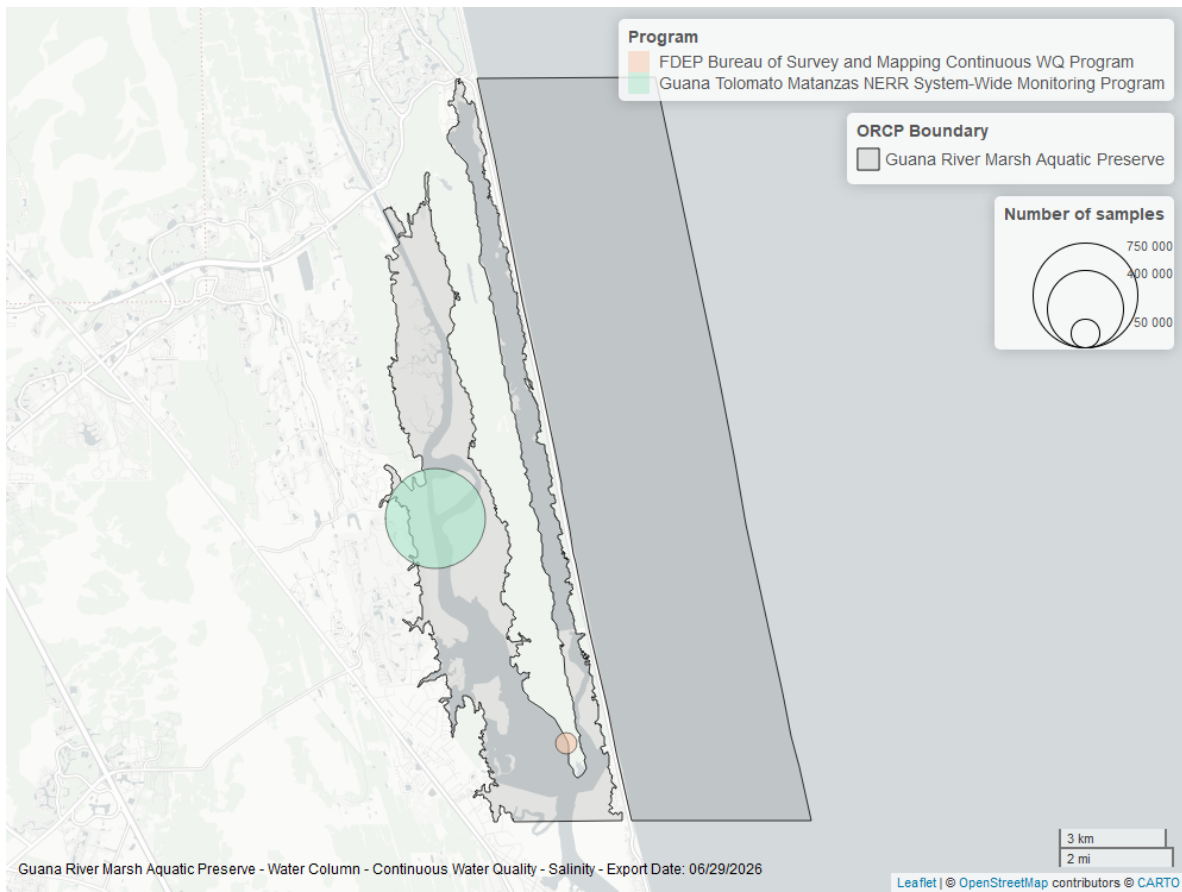


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

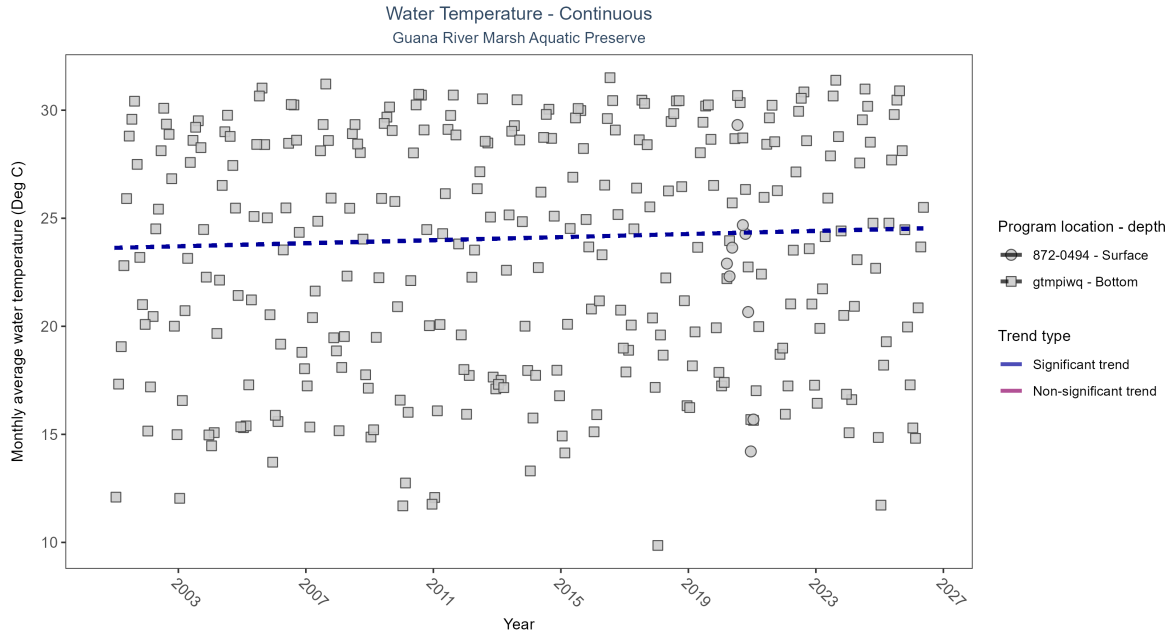


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: At one program location, monthly average water temperature increased by 0.04Deg C per year. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmptwq	Increasing trend	763299	26	2001 - 2026	24.20	0.18	23.63	0.04	0
872-0494	Insufficient data	35473	2	2020 - 2021	22.34	-	-	-	-

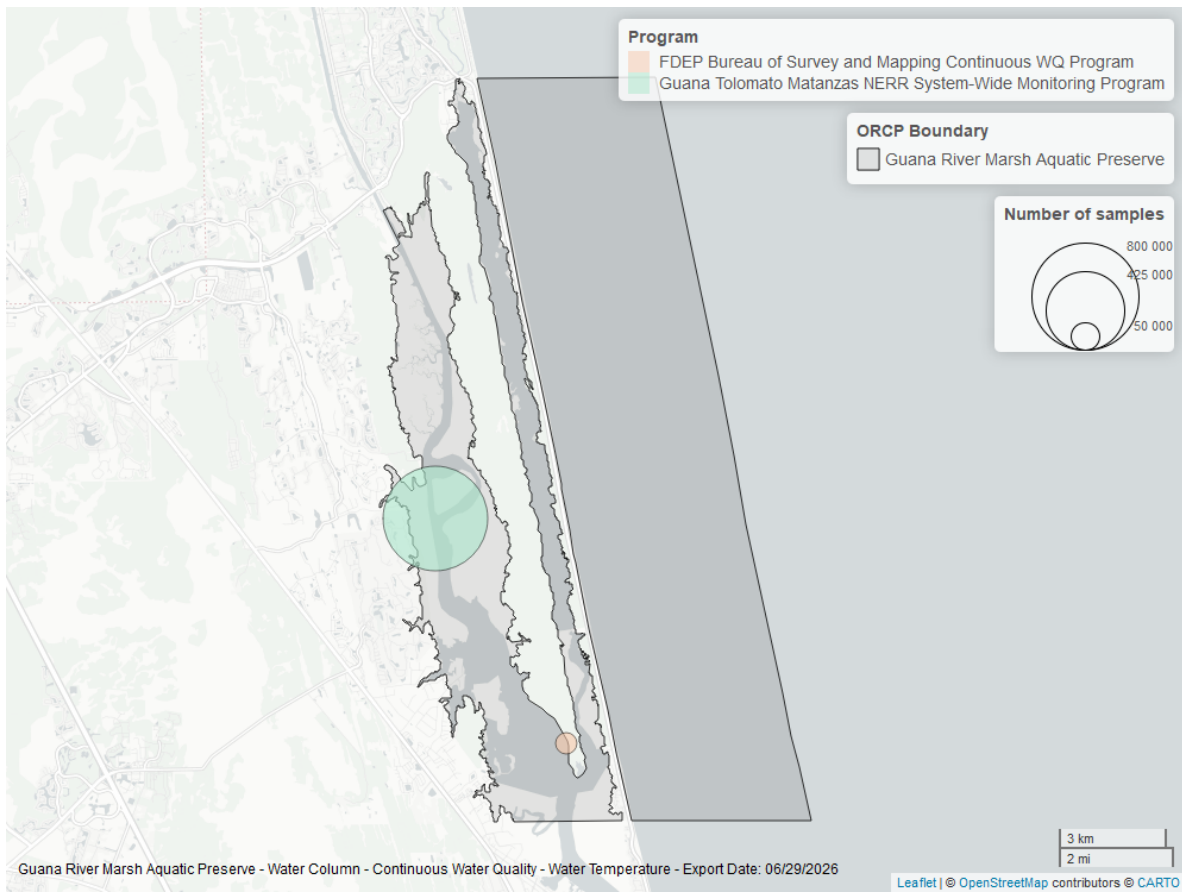


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

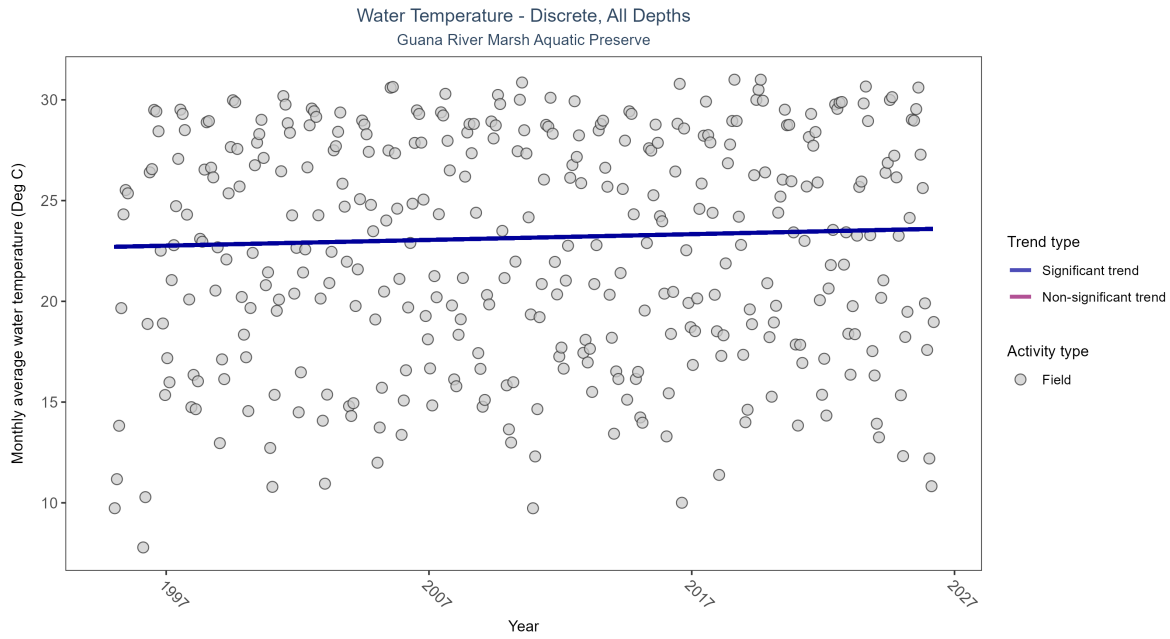


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 8: Monthly average water temperature increased by 0.03Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	8423	32	1995 - 2026	23.1	0.10907	22.70534	0.02856	0.0029

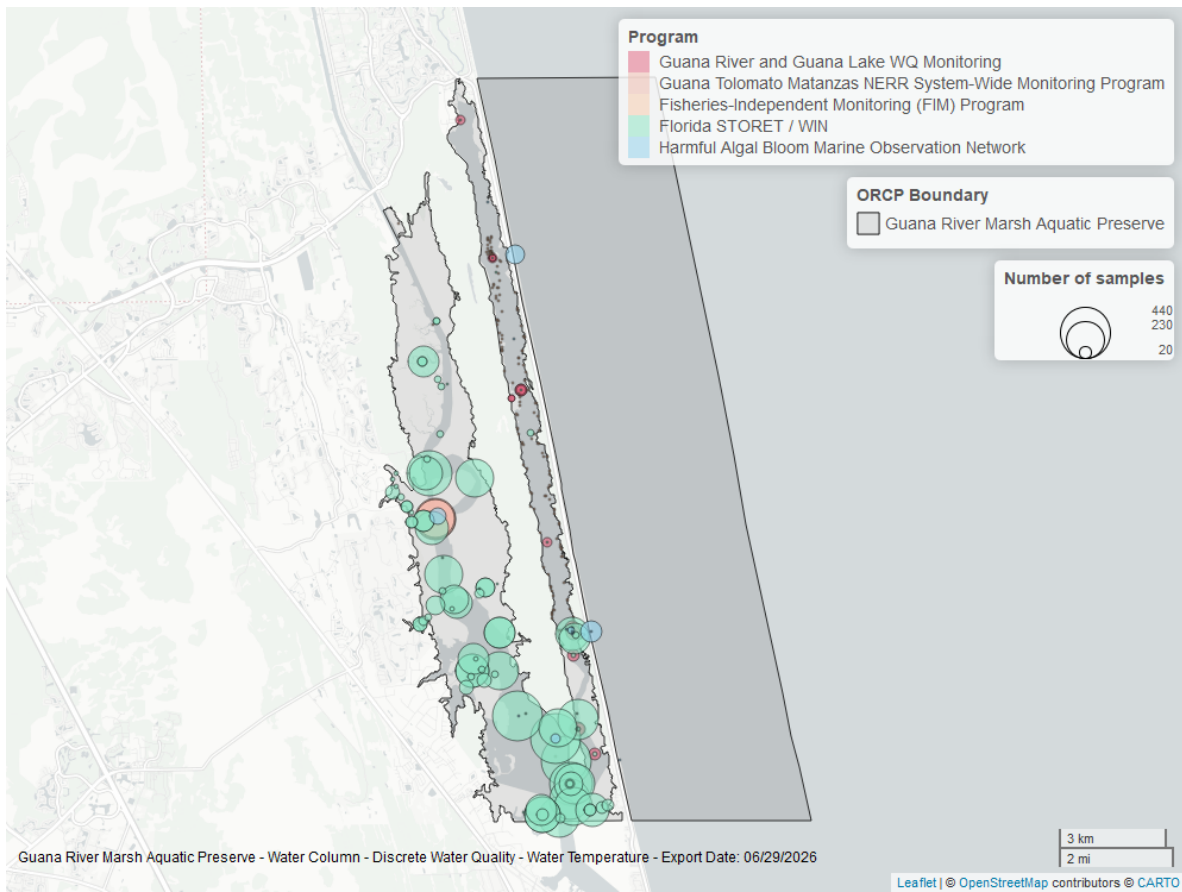


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

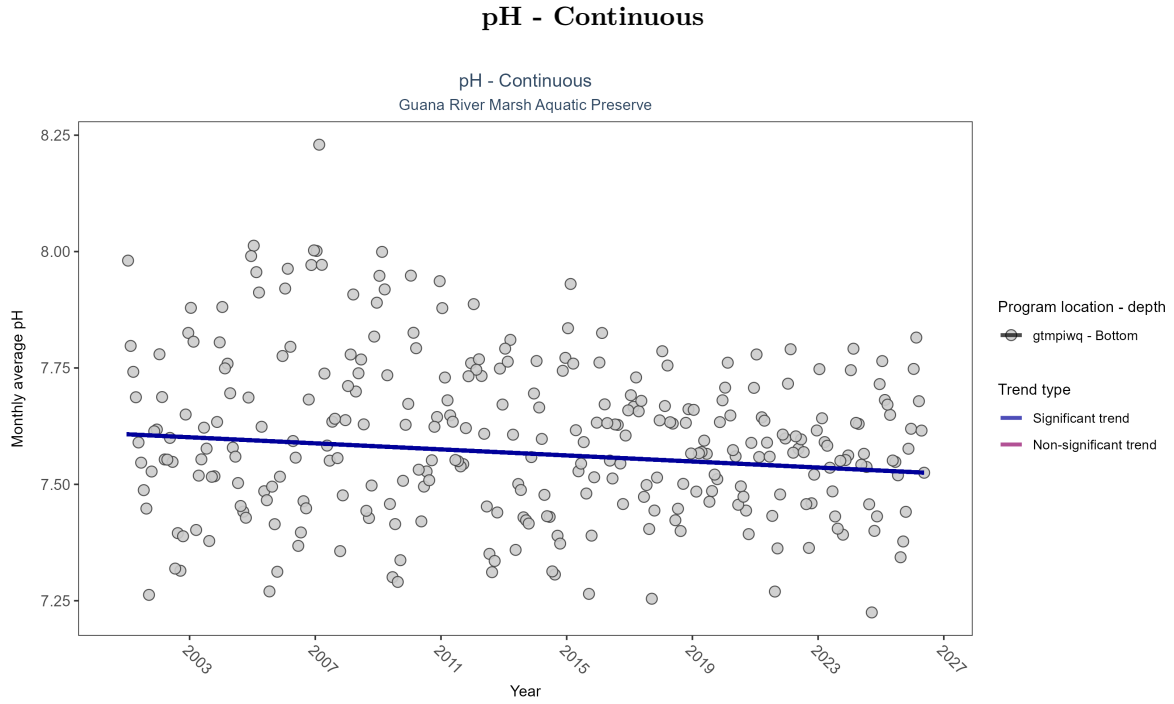


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: At one program location, monthly average pH decreased by less than 0.01 pH units per year.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Decreasing trend	705937	26	2001 - 2026	7.6	-0.17	7.61	0	0

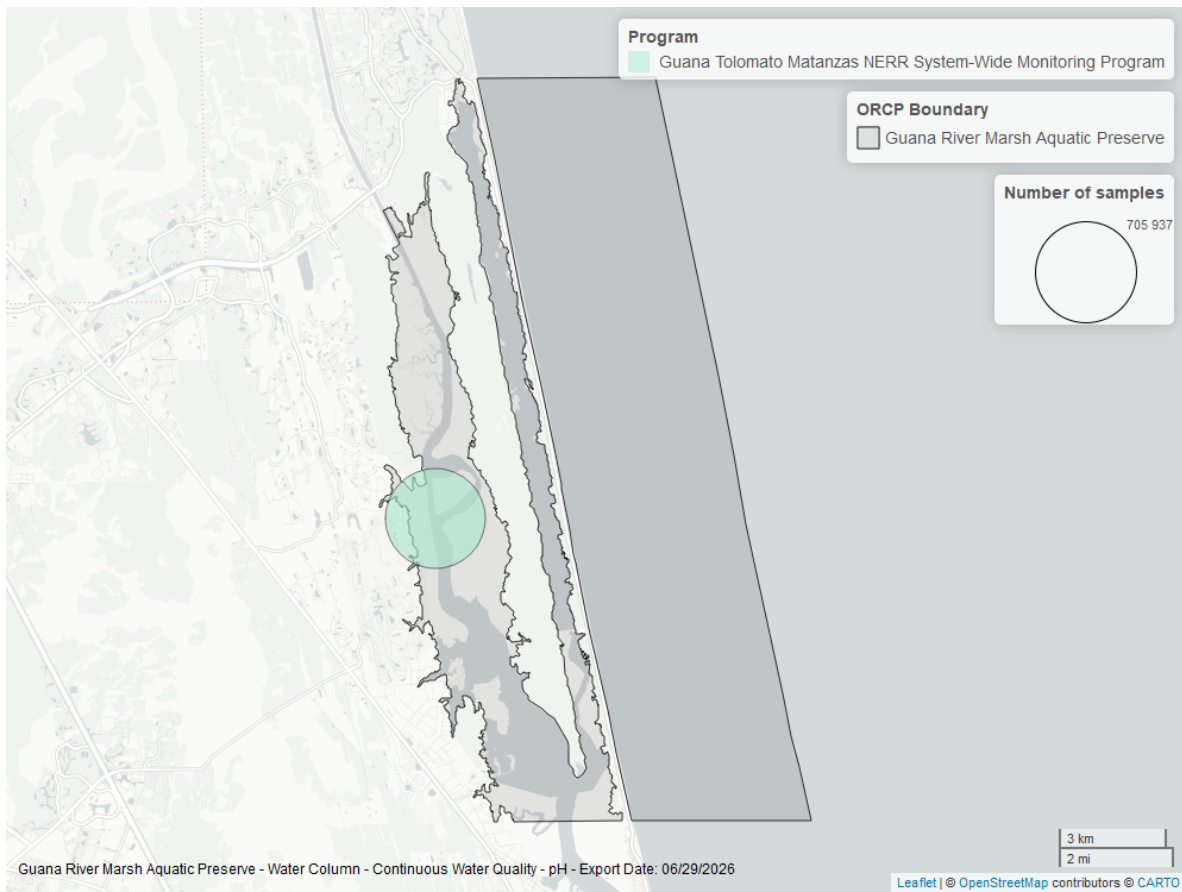


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

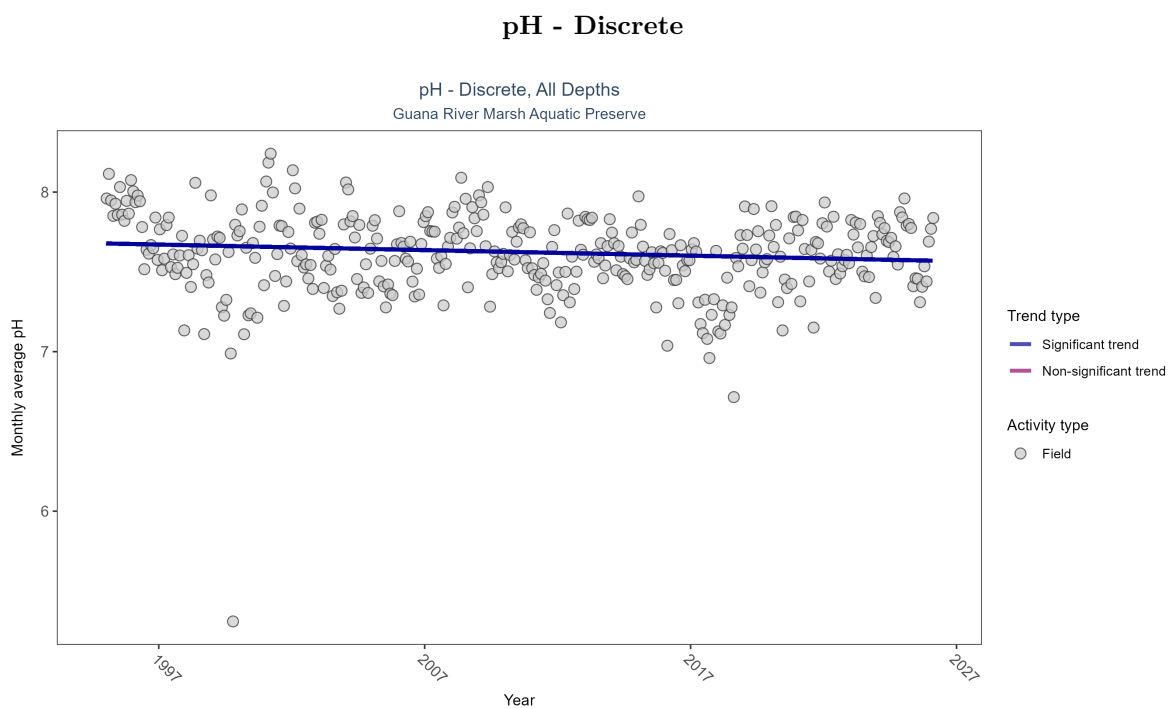


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 10: Monthly average pH decreased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	6283	32	1995 - 2026	7.7	-0.11196	7.67802	-0.00347	0.0022

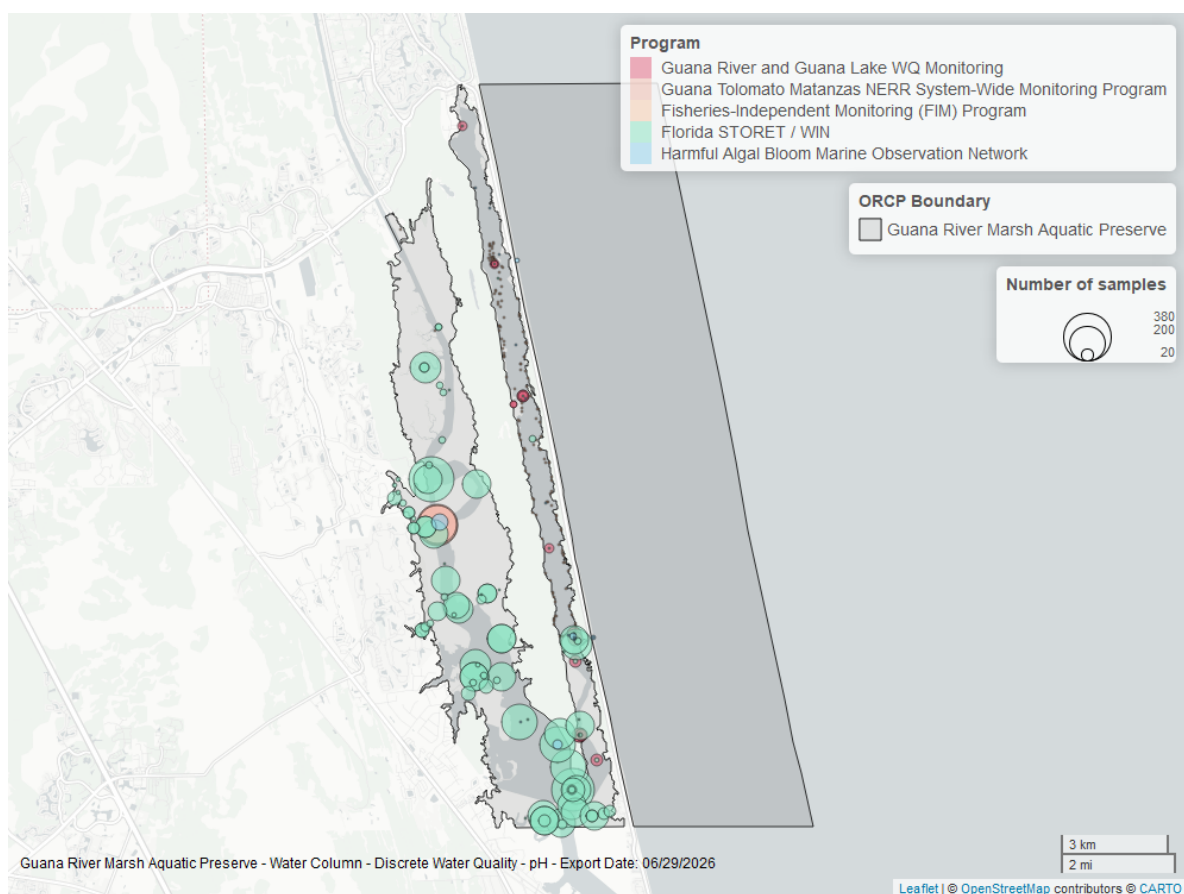


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

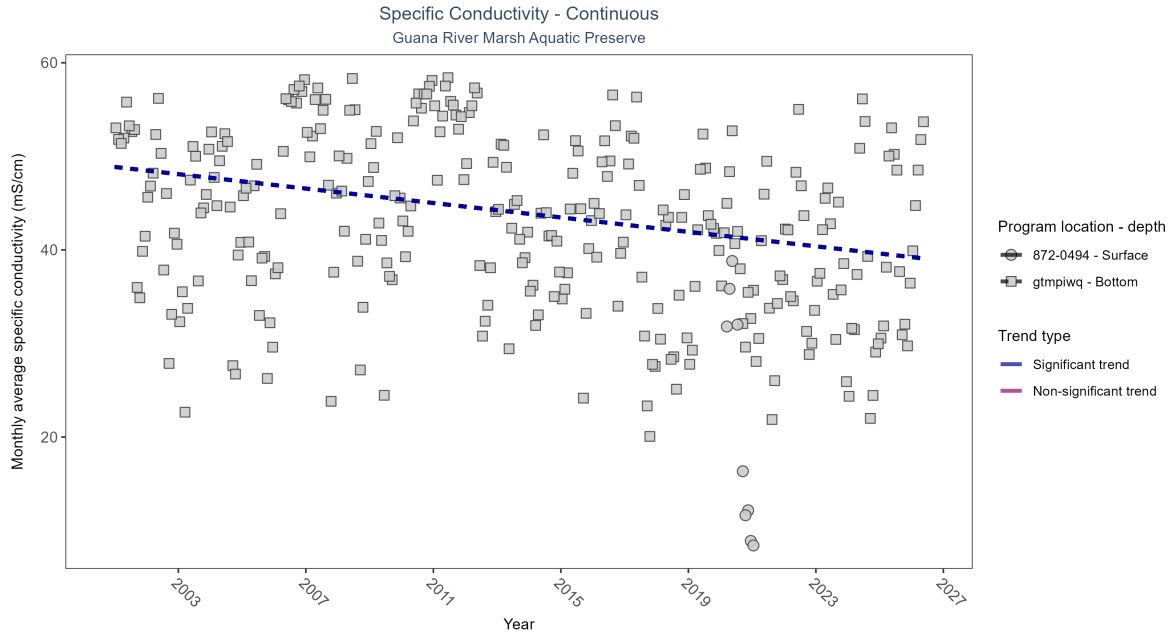


Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: At one program location, monthly average specific conductivity decreased by 0.39 mS/cm per year. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmptwq	Decreasing trend	713006	26	2001 - 2026	43.23	-0.25	48.87	-0.39	0
872-0494	Insufficient data	34929	2	2020 - 2021	14.00	-	-	-	-

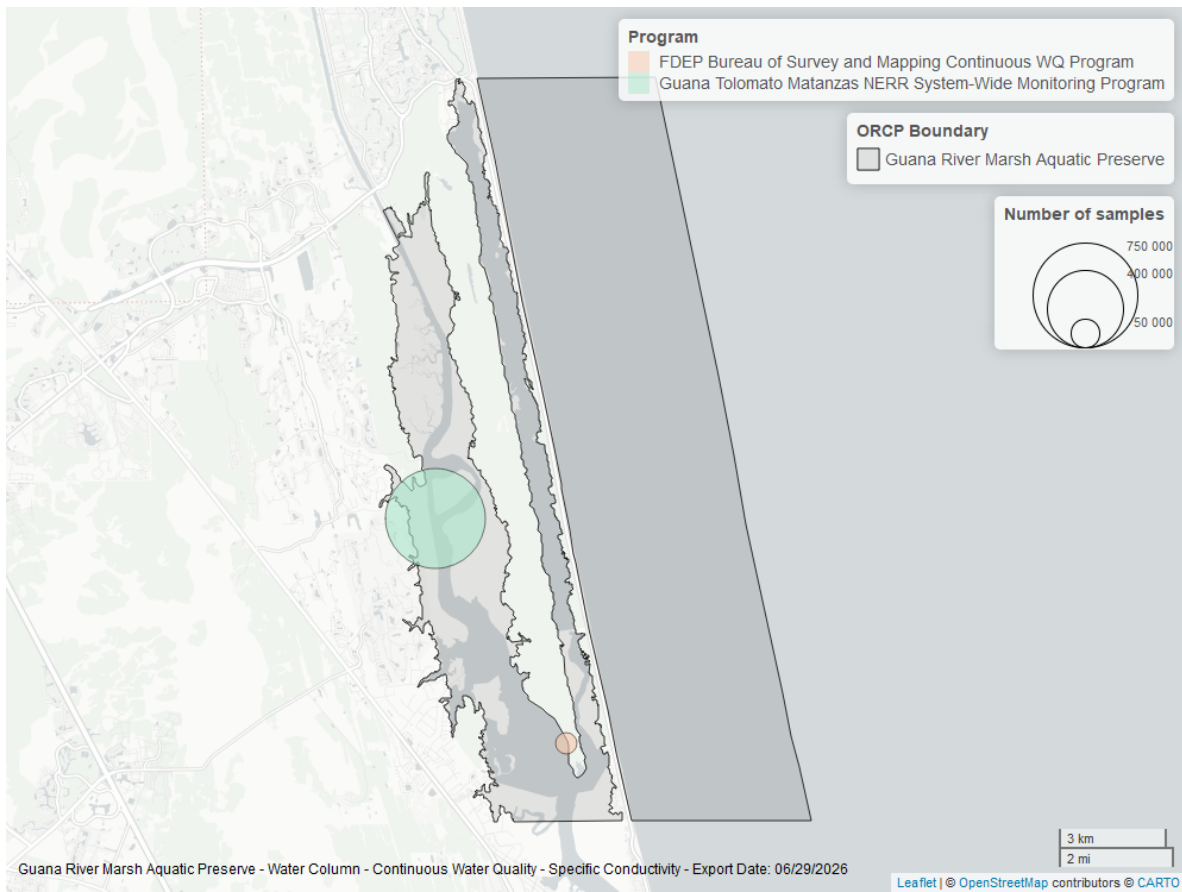


Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

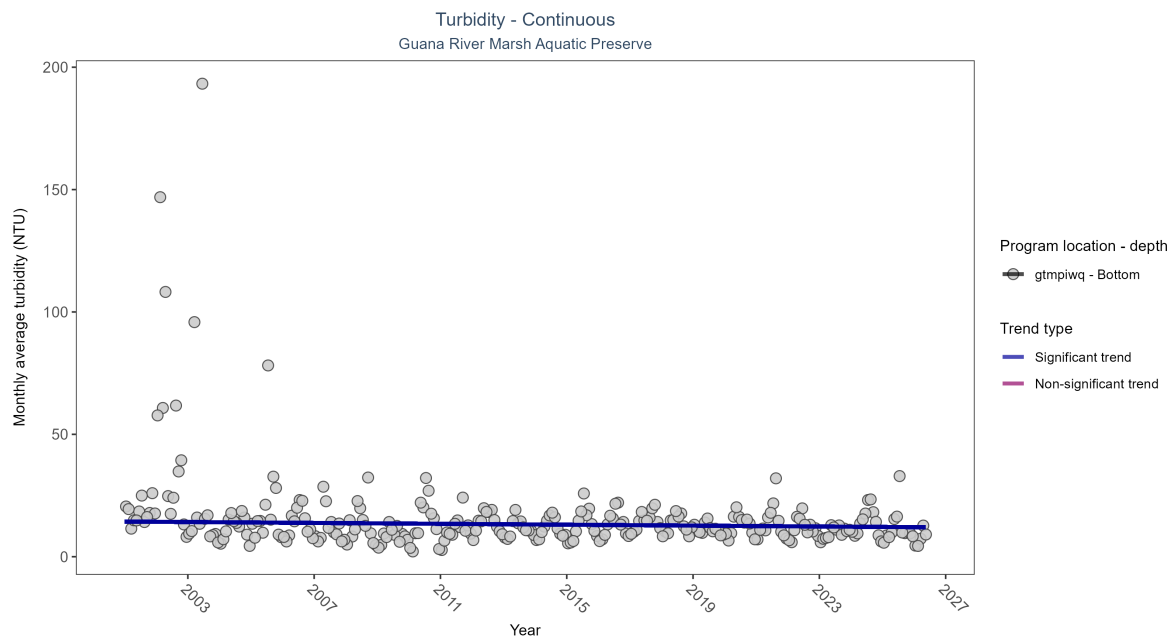


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 12: At one program location, monthly average turbidity decreased by 0.09 NTU per year.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmpiwq	Decreasing trend	689004	26	2001 - 2026	10	-0.13	14.35	-0.09	0

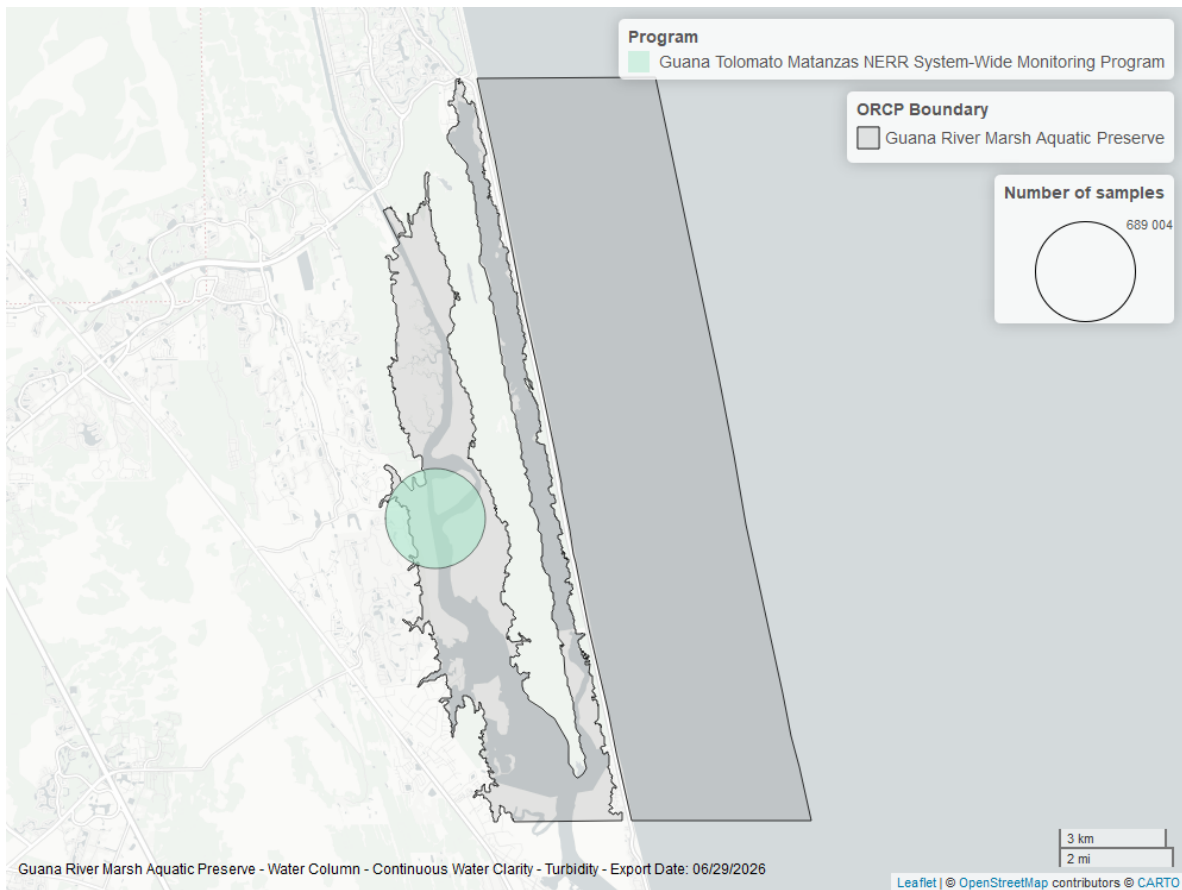


Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

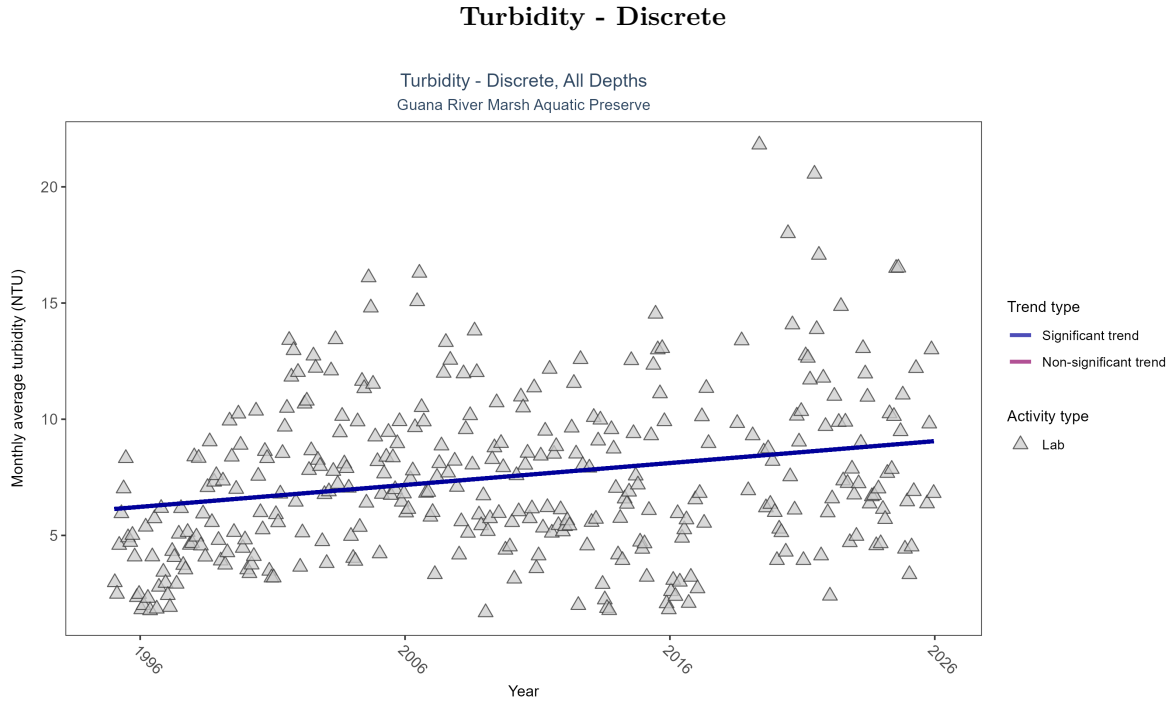


Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 13: Monthly average turbidity increased by 0.09 NTU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	5037	31	1995 - 2025	5.2	0.21145	6.13529	0.09426	0

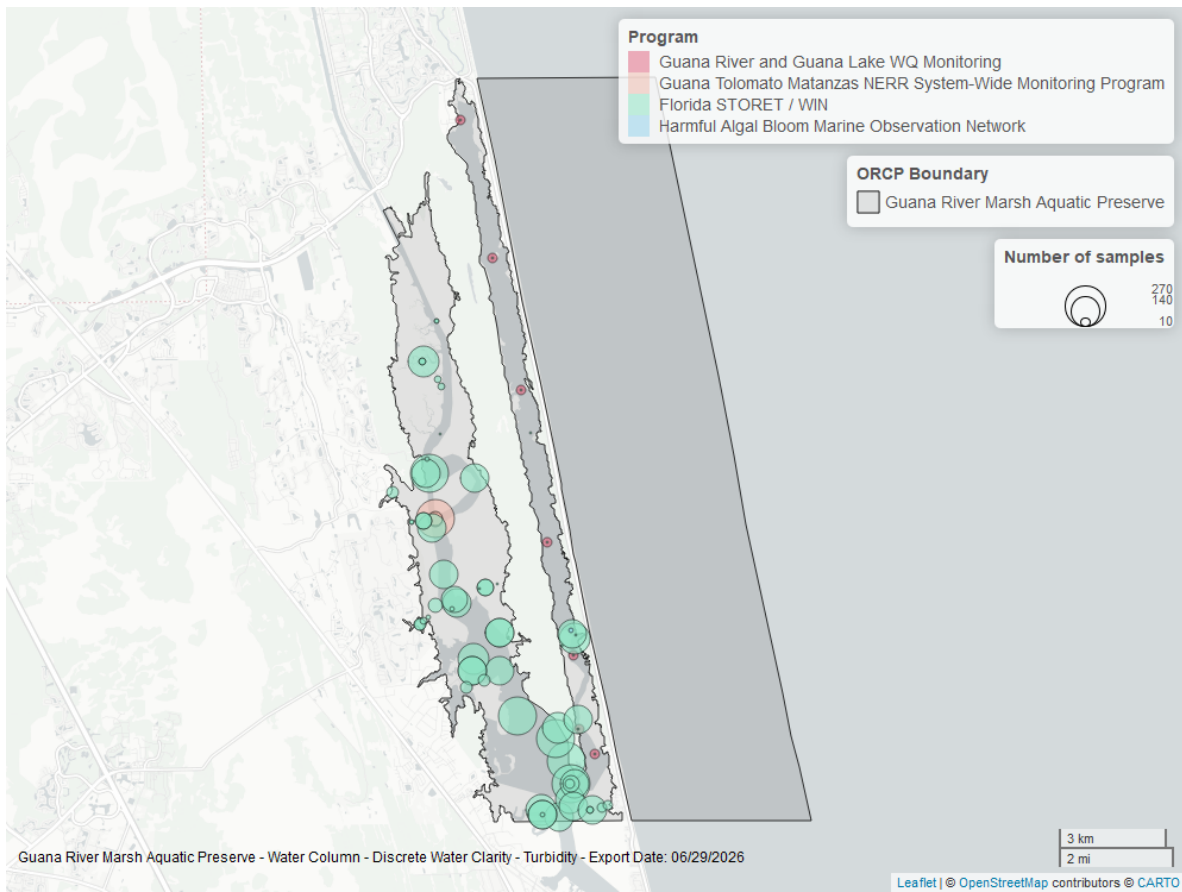


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

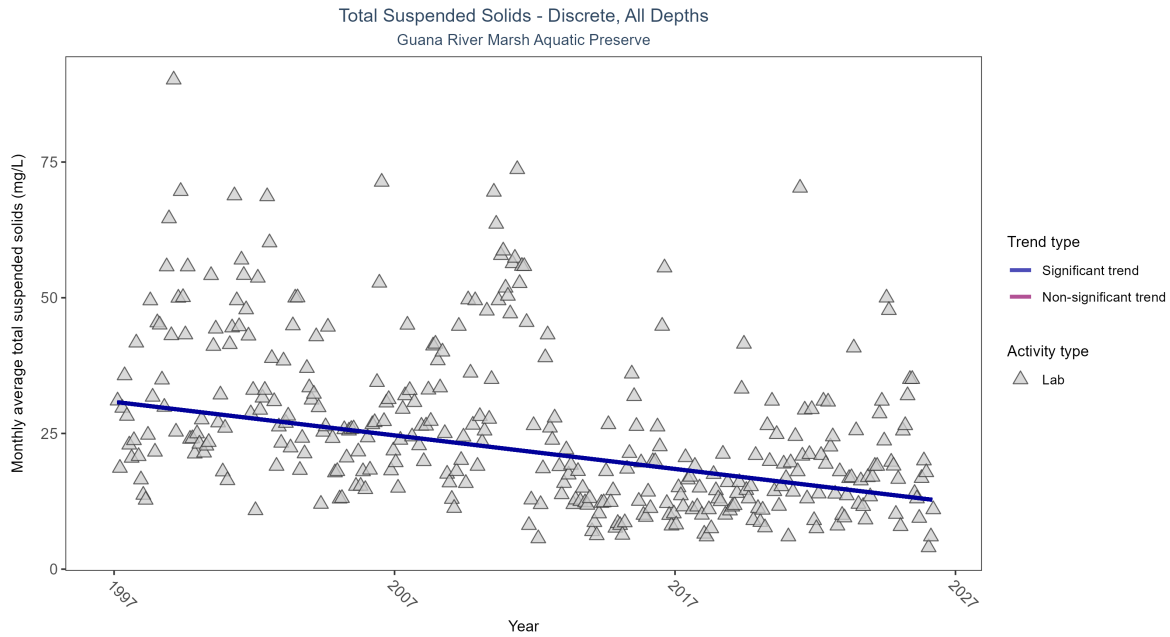


Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 14: Monthly average total suspended solids decreased by 0.62 mg/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	1753	30	1997 - 2026	22	-0.33847	30.82747	-0.61842	0

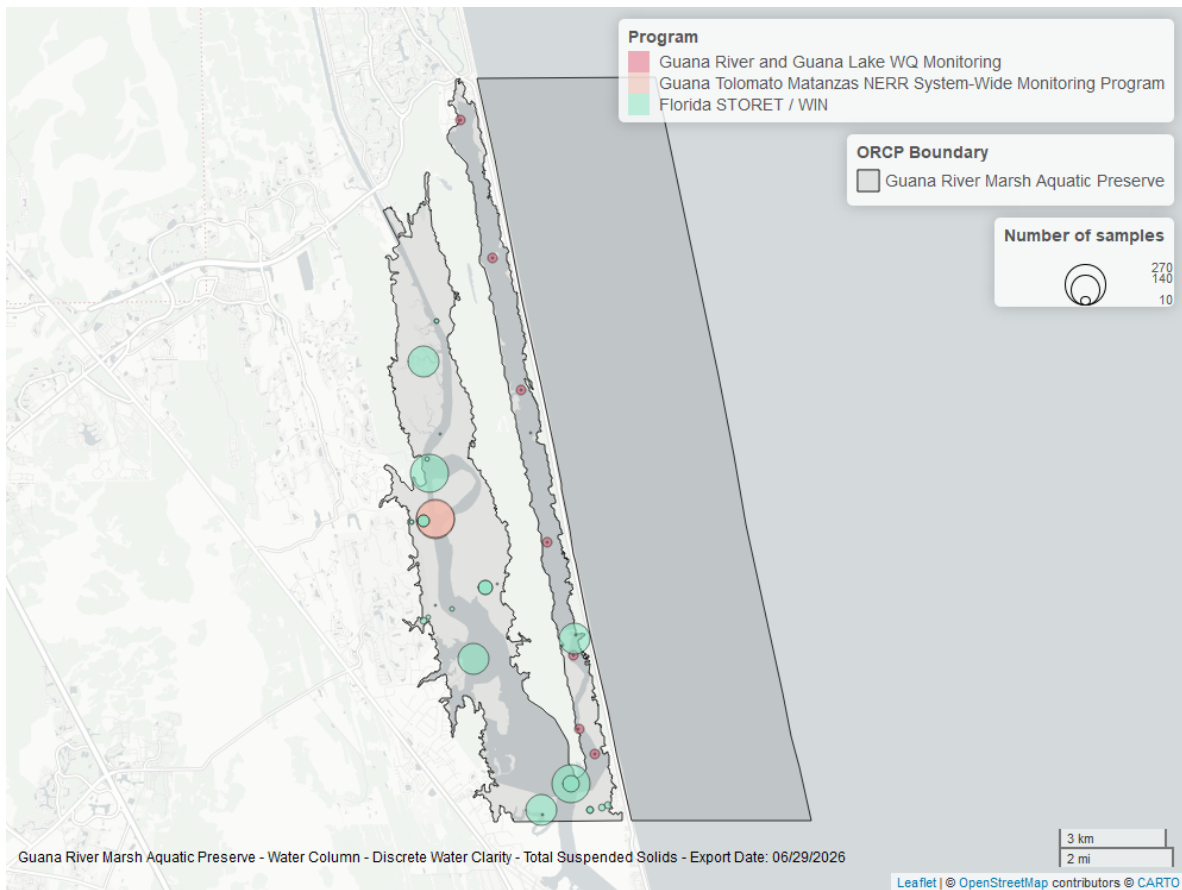


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

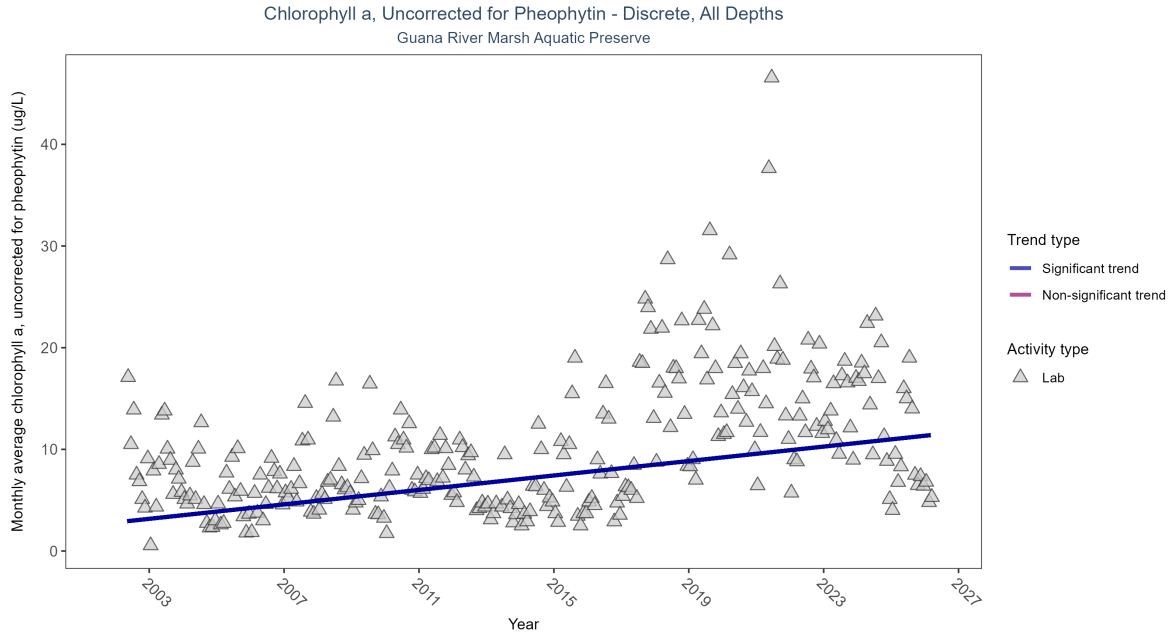


Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 15: Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.36 ug/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1363	25	2002 - 2026	7.8	0.38139	2.81288	0.35522	0

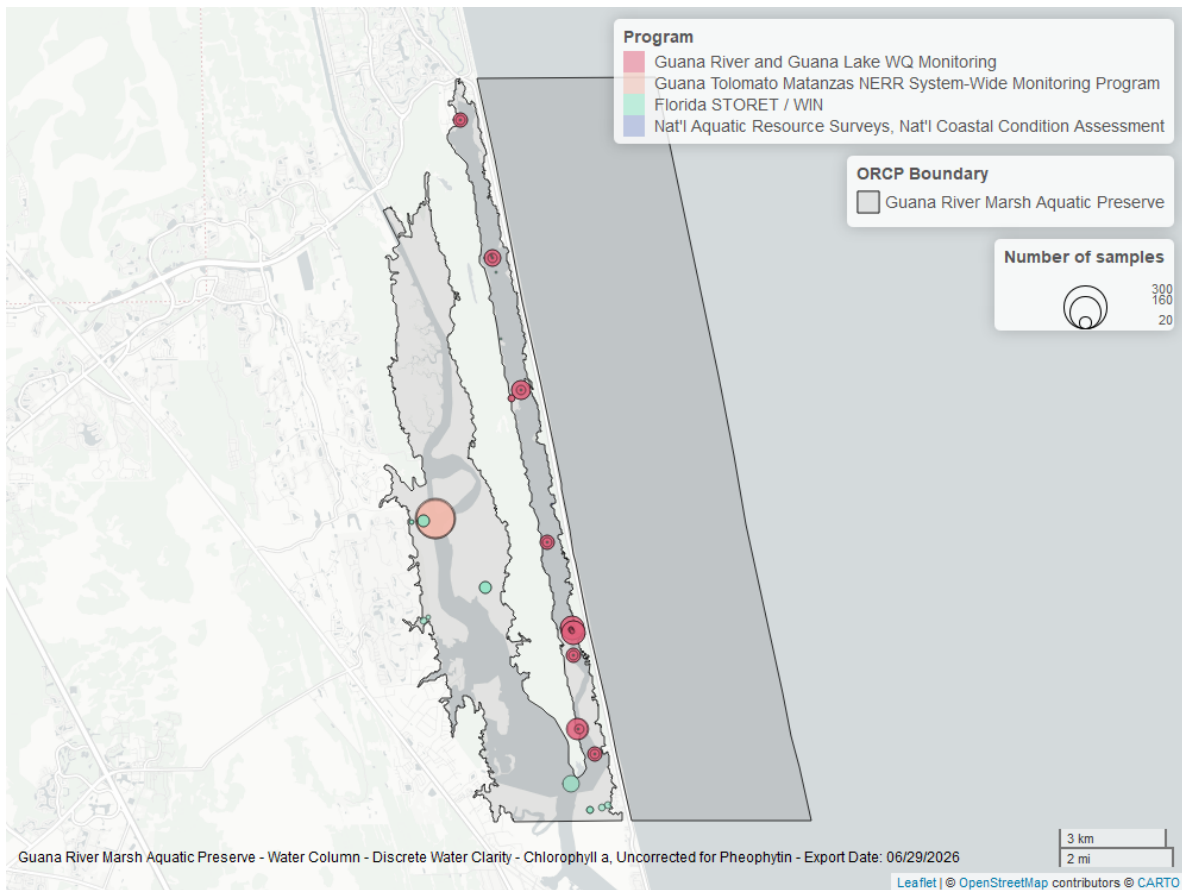


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

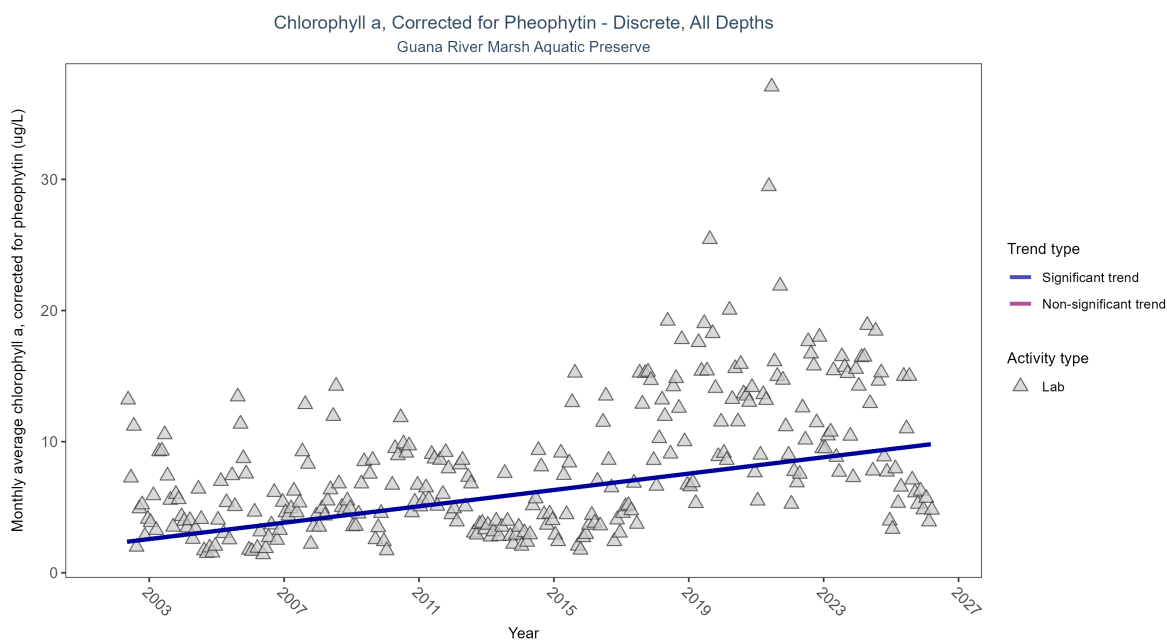


Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 16: Monthly average chlorophyll a, corrected for pheophytin, increased by 0.31 ug/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1727	25	2002 - 2026	5.5	0.39333	2.26052	0.31199	0



Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

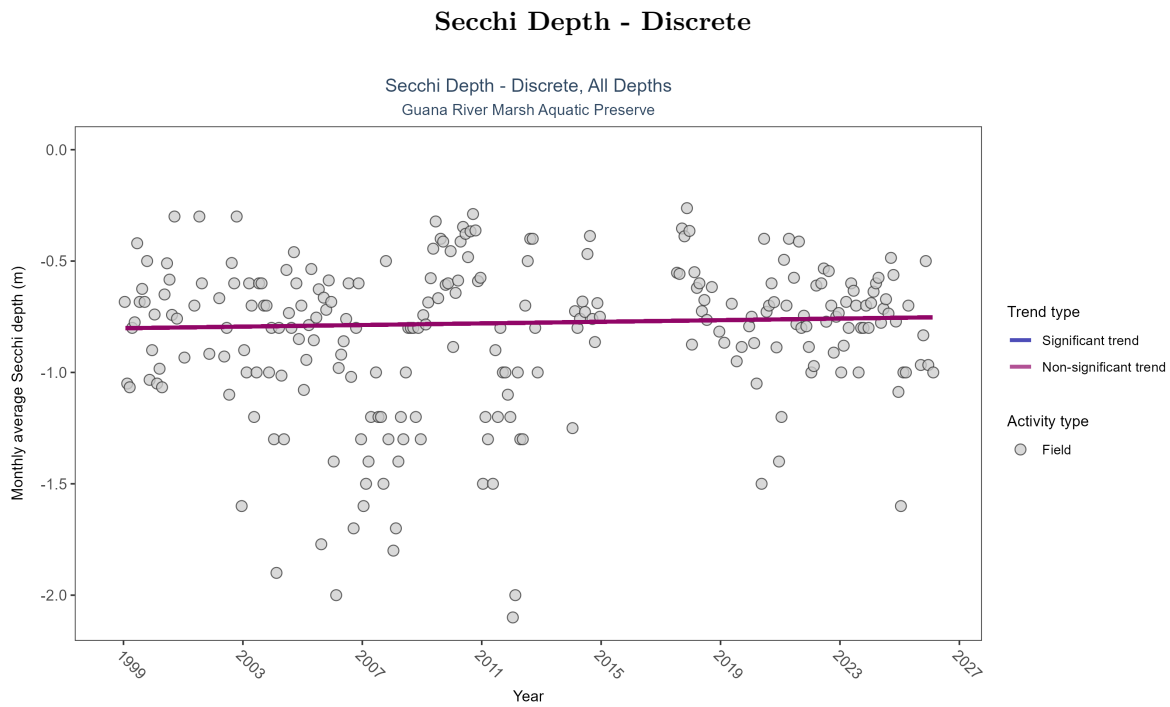


Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 17: Secchi depth showed no detectable trend between 1999 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	1424	25	1999 - 2026	-0.7	0.04647	-0.80158	0.00181	0.3731

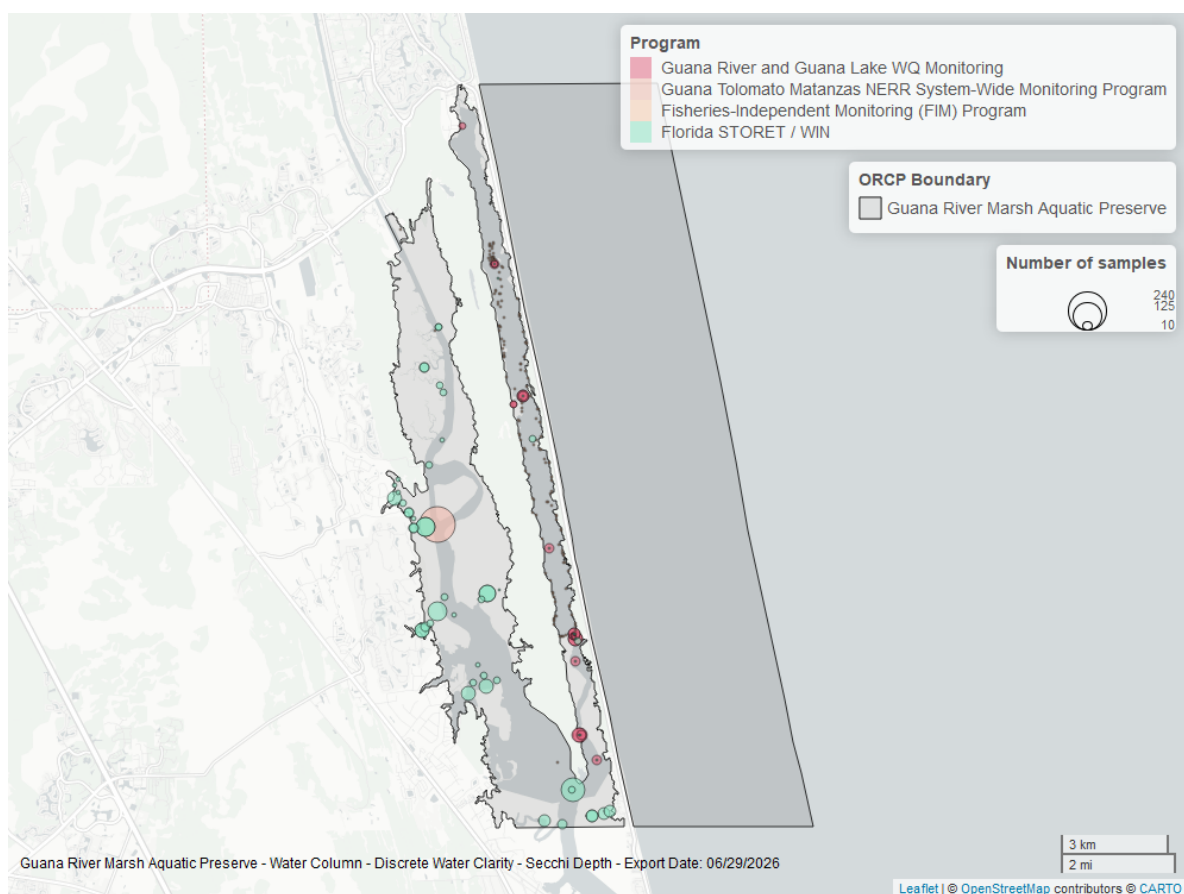


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

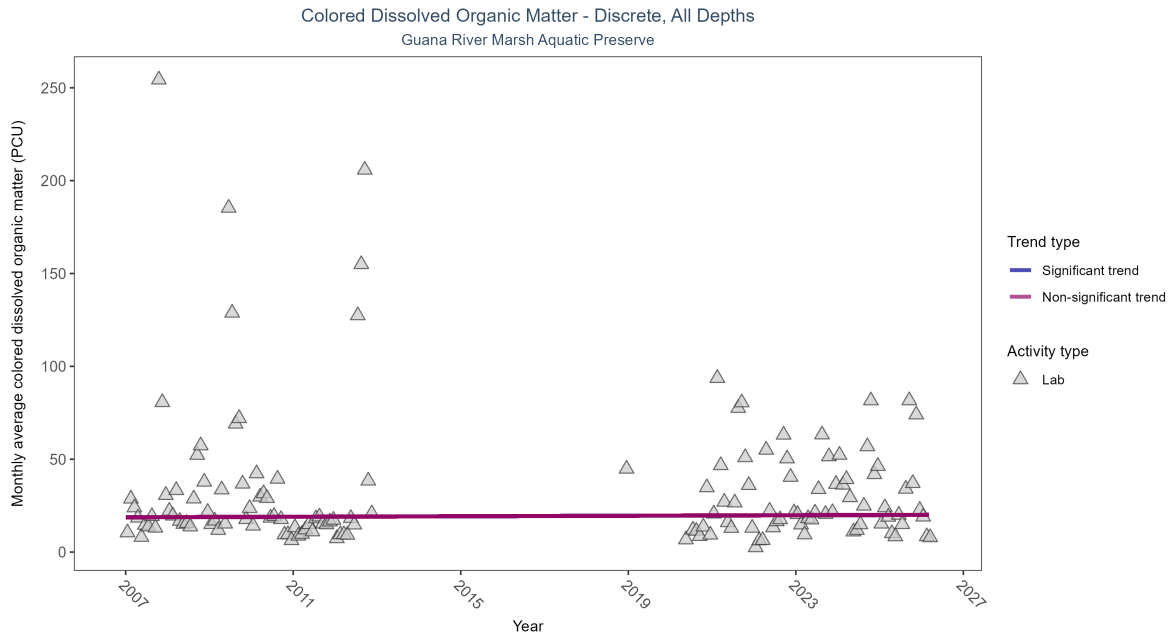


Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 18: Colored dissolved organic matter showed no detectable trend between 2007 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	282	14	2007 - 2026	19.7619	0.0233	18.73976	0.06858	0.7164

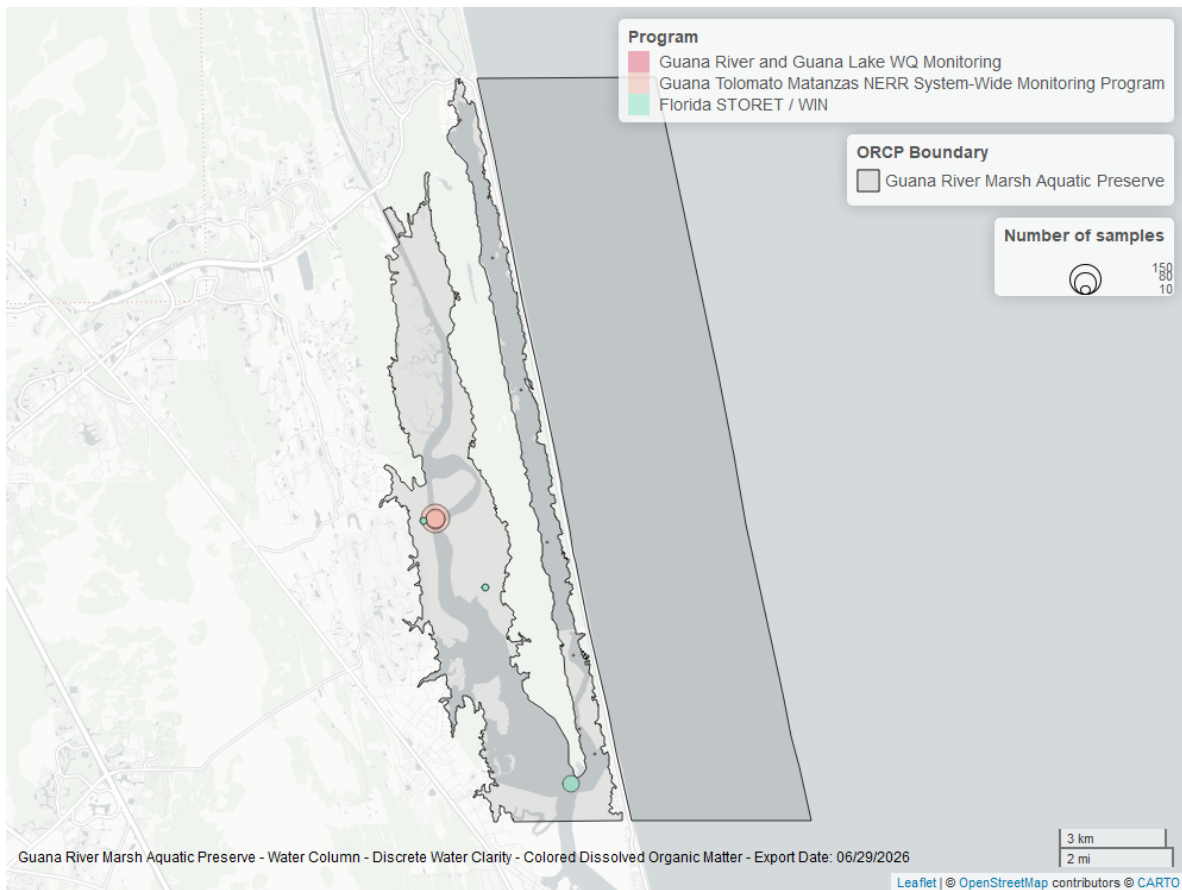


Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.